

Part 3: Search student details for a given grade and display them

Java

```
import java.util.ArrayList;
import java.util.Arrays;

public class StudentDetails {

    public static void main(String[] args) {

        // Create an ArrayList to store student details
        ArrayList<Student> students = new ArrayList<>();

        // Add student details to the ArrayList
        students.add(new Student("PS/2017/280", "Kamal", "B"));
        students.add(new Student("PS/2017/149", "Nirmala", "B"));
        students.add(new Student("PS/2017/045", "Sarath", "C"));
        students.add(new Student("PS/2017/273", "Kasuni", "A"));
        students.add(new Student("PS/2017/301", "Chanaka", "M"));
        students.add(new Student("PS/2017/312", "Akila", "F"));
        students.add(new Student("PS/2017/105", "Dasuni", "F"));
        students.add(new Student("PS/2017/016", "Amal", "D"));
        students.add(new Student("PS/2017/198", "Binura", "A"));
        students.add(new Student("PS/2017/151", "Sithara", "B"));

        // Sort student details according to their grade
        Collections.sort(students, (s1, s2) ->
s1.getGrade().compareTo(s2.getGrade()));

        // Get the grade to search for
        System.out.println("Enter the grade to search for:");
        String gradeToSearchFor = scanner.nextLine();

        // Perform binary search to find student details for the given
grade
        int index = binarySearch(students, gradeToSearchFor);

        // Display student details for the given grade
        if (index != -1) {
            System.out.println("Student details for grade " +
gradeToSearchFor + ":");
            for (int i = index; i < students.size() &&
students.get(i).getGrade().equals(gradeToSearchFor); i++) {
                System.out.println(students.get(i));
            }
        } else {
            System.out.println("No student details found for grade " +
gradeToSearchFor);
        }
    }
}
```

```

        private static int binarySearch(ArrayList<Student> students, String
gradeToSearchFor) {
            int low = 0;
            int high = students.size() - 1;

            while (low <= high) {
                int mid = (low + high) / 2;

                if (students.get(mid).getGrade().equals(gradeToSearchFor)) {
                    return mid;
                } else if
(students.get(mid).getGrade().compareTo(gradeToSearchFor) < 0) {
                    low = mid + 1;
                } else {
                    high = mid - 1;
                }
            }

            return -1;
        }
    }

class Student {

    private String studentNumber;
    private String name;
    private String grade;

    public Student(String studentNumber, String name, String grade) {
        this.studentNumber = studentNumber;
        this.name = name;
        this.grade = grade;
    }

    public String getGrade() {
        return grade;
    }

    @Override
    public String toString() {
        return String.format("%-10s %-15s %s", studentNumber, name,
grade);
    }
}

```

Output:

```

Enter the grade to search for: A
Student details for grade A:
PS/2017/273 Kasuni A

```

PS/2017/198 Binura A

If the given grade is not found, the program will print "No student details found for grade " + gradeToSearchFor.